

CISC 3667: Game Design and Development

In-Class Activity 5: Balancing Games

Assigned: Thursday, October 1

Submission: Brightspace by 11:59 PM (Individual Submission)

Objective

Game balance involves adjusting a game's mechanics, rules, and numerical values to create the intended player experience. A game may be technically functional but still suffer from problems such as dominant strategies, unfair advantages, inappropriate difficulty, or meaningless choices.

In this activity, you will analyze an intentionally unbalanced game, identify problems with its design, investigate its mechanics, and propose adjustments to improve the gameplay experience. You will then evaluate your redesigned game using hypothetical gameplay scenarios to determine whether your adjustments address the original balancing problems or introduce new ones.

You may work with your group to discuss ideas, analyze the game, and explore possible solutions. However, **each student must develop their own analysis, justify their proposed changes, and submit their own completed activity.**

You are not required to physically build, program, or playtest the game.

By completing this activity, you should be able to:

- Identify and explain balancing problems involving fairness, challenge, and meaningful choices.
- Recognize dominant strategies and analyze how they affect gameplay.
- Use numerical analysis to investigate the effectiveness of different game mechanics.
- Apply triangularity to design meaningful risk-and-reward decisions.
- Propose and justify balancing adjustments to an existing game.
- Evaluate how different gameplay situations affect player decisions and the effectiveness of game mechanics.

Part 1: Analyze an Unbalanced Game

Consider the following game, Race for the Treasure. Although the game is playable, its mechanics contain several intentional balancing problems.

Your first task is to examine the original rules and identify those problems before attempting to redesign the game.

Race for the Treasure

Objective: Be the first player to reach or pass space 20.

Setup:

- Two players play.
- Both players begin on space 0 of a 20-space track.
- Player 1 always takes the first turn.
- Players alternate turns until someone wins.

Gameplay:

On each turn, a player must select exactly one of the following actions.

Action	Effect
Safe Move	Advance 3 spaces automatically.
Risky Move	Roll a six-sided die. Advance 7 spaces if you roll 4–6. Otherwise, remain in place.
Power Move	Advance 5 spaces automatically. You may use this action twice per game .

Additional Rules:

- Players may choose any action for which they meet the requirements.
- Each player may use Power Move at most twice per game. Safe Move and Risky Move have no usage limits.
- Players cannot interfere with each other.
- The first player to reach or pass space 20 immediately wins.

Questions

Analyze the original game using the concepts discussed in class.

1. **Fairness:** Is the game symmetrical? Does giving both players identical actions necessarily guarantee a fair experience? Consider the starting conditions, available actions, and turn order.
2. **Meaningful Choices:** Compare the three available actions. Does each action offer a meaningful strategic advantage? Explain whether players have a reason to select different actions depending on the situation.
3. **Dominant Strategies:** Does the game contain a dominant strategy? If so, identify it and explain how it affects player decision-making and the overall gameplay experience.
4. **Challenge:** Evaluate the game's level of challenge. Does the current design require meaningful decisions or strategic thinking? Explain how the available mechanics influence the difficulty of winning.
5. **Triangularity:** Examine the relationship between risk, reward, and certainty in the original game. Does the Risky Move offer a meaningful alternative to the other actions? Explain your reasoning.
6. **Identifying Balancing Problems:** Based on your analysis, **identify three potential balancing concerns or trade-offs**. For each problem, explain which mechanic or rule causes it and how it affects the gameplay experience.

Part 2: Investigate the Game Mechanics

Before redesigning the game, investigate how the numerical values of its existing mechanics affect player decisions.

A useful technique in game balancing is to compare the expected outcomes of different actions.

The expected value of an action represents its average outcome over many attempts. It helps designers identify numerical advantages and assess whether different actions provide appropriate rewards.

For an action with multiple possible outcomes, its expected value can be calculated using:

$$E(X) = \sum x \cdot P(x)$$

Where:

- $E(X)$ represents the expected value.
- $P(x)$ represents the probability of an outcome.
- x represents the numerical value of that outcome.

However, expected value alone does not determine whether an action is balanced. Designers must also consider risk, certainty, the current game state, and the player's objectives.

Questions

1. Calculate the Expected Movement

Calculate the expected number of spaces a player advances when selecting each of the original game's three actions. Show your calculations and complete the following table.

Action	Possible Outcomes	Probability of Each Outcome	Expected Movement
Safe Move			
Risky Move			
Power Move			

For the Risky Move, remember that the player advances 7 spaces when rolling 4, 5, or 6 and remains in place when rolling 1, 2, or 3.

2. Compare the Actions: Based on your calculations, which action provides the highest expected movement? Does this automatically make it the most effective action in every possible situation? Explain your reasoning.

3. Risk Versus Reward: Compare the Risky Move and the Power Move. Does the potential reward of the Risky Move justify its uncertainty? Explain why a player might or might not choose it over the Power Move.

4. Numerical Balance and Meaningful Choices

Suppose you could only modify the numerical values of the three actions without introducing additional rules or mechanics.

- What values would you change to make the actions more competitive?

Propose **at least one numerical adjustment** and explain how it might influence player decisions.

These preliminary adjustments do not have to be the same as the changes you develop in Part 3.

Part 3: Redesign and Balance the Game

Now that you have identified and investigated the original game's balancing problems, your task is to redesign its mechanics.

Your goal is to create a game in which players have meaningful choices, different actions provide appropriate advantages and disadvantages, and the rules support a fair and challenging gameplay experience.

You may modify numerical values, introduce restrictions, adjust existing mechanics, or add a limited number of new rules.

However, your redesigned game must preserve the following elements:

- **The original objective of reaching or passing space 20.**
- **A two-player race using the same 20-space track.**
- **Three available actions: Safe Move, Risky Move, and Power Move.**

You may change how these actions function, but you cannot remove or replace them entirely. Your redesigned game must address the following areas:

A. Fairness

Evaluate the original game's starting conditions and determine whether either player receives an unintended advantage.

- Propose at least **one adjustment** intended to improve fairness. You may consider modifying turn order, introducing compensating advantages, or adjusting other mechanics.
- Explain why your proposed change should **improve fairness and whether it introduces any new balancing concerns.**

B. Meaningful Choices

Redesign the three actions so that each offers a meaningful strategic option.

Players should have reasons to select different actions depending on the circumstances. Avoid creating a situation in which one action is consistently superior to every alternative.

Complete the following table.

Action	Revised Effect	Advantage	Disadvantage
Safe Move			
Risky Move			
Power Move			

For each action, explain a situation in which a player might reasonably choose it over the alternatives.

Your explanations should demonstrate how the revised mechanics create meaningful decisions rather than simply providing different numerical values.

C. Challenge

Consider how your redesigned mechanics affect the difficulty of winning.

Your game should provide an appropriate level of challenge without making success entirely dependent on chance or creating unnecessary frustration.

Identify **one mechanic** in your redesigned game that contributes to its challenge.

Explain:

- How the mechanic works.
- What decisions players must make in response to it.
- How it contributes to the overall gameplay experience.

Consider whether your changes allow players to make strategic decisions throughout the game, including when they are ahead or behind.

D. Complete Revised Rules

Write a complete set of rules for your redesigned game.

Your rules should clearly explain the **starting conditions, available actions, any new restrictions or mechanics, turn structure, and win condition.**

Another student should be able to understand how your redesigned game works without requiring additional explanations.

Part 4: Triangularity and Risk-Reward Analysis

One important aspect of game balance is **triangularity**, which involves creating meaningful decisions between **risk**, **reward**, and **certainty**. Players may choose a relatively safe action that provides a smaller but more predictable reward or pursue a riskier action that offers a potentially greater reward.

Your redesigned game must include **at least one meaningful decision** involving this relationship.

Questions

1. Identify **one relatively low-risk action** and **one relatively high-risk action** in your redesigned game.

	Low-Risk Choice	High-Risk Choice
Action		
Potential reward		
Risk or disadvantage		
Certainty of receiving the reward		
Situation where this choice would be useful		

2. Evaluate the Relationship: Explain how your revised mechanics **balance risk, reward, and certainty**. Why might a player reasonably **choose the riskier action instead of the safer alternative**?

3. Meaningful Decisions: Describe a gameplay situation in which a player would have **difficulty deciding between the two actions**. Explain **what makes the decision meaningful** and **how the player's circumstances might influence their choice**.

Part 5: Evaluate Your Redesigned Game

Balancing a game requires designers to consider how their mechanics function in different situations. A mechanic that appears balanced under one set of circumstances may become disproportionately powerful or ineffective when the game state changes.

In this final section, you will evaluate your redesigned game using three hypothetical gameplay scenarios. You are not required to physically playtest the game. Instead, use your revised rules, numerical analysis, and understanding of game balance to predict how players might behave.

For each scenario, explain which actions would be available, which decisions players might make, and whether your redesigned mechanics support meaningful choices. If your revised actions depend on resources, remaining uses, or other conditions, state your assumptions for each scenario.

- **Scenario A: Approaching the Finish Line:**

- **Both players are tied on space 17. It is Player 1's turn.**
- Using your redesigned rules:
 - What actions are available to Player 1, and what are their possible outcomes?
 - Which action would you consider selecting in this situation? Explain your reasoning.
 - Does your design provide multiple reasonable options, or does one action become clearly preferable? Explain.

- **Scenario B: Falling Behind:**

- **Player 1 is on space 15, while Player 2 is on space 8. It is Player 2's turn.**
- Using your redesigned rules:
 - What options does Player 2 have for attempting to recover?
 - Does your game provide meaningful decisions for a player who is significantly behind?
 - Would you introduce any additional balancing adjustments to address this situation? Explain why or why not.

- **Scenario C: Repeated Strategies:**

- Imagine that a player repeatedly selects the same action throughout the game.
- Under your revised rules, could repeatedly selecting one action become a dominant strategy? Explain.
- Identify a situation in which choosing a different action would be beneficial.
- What additional adjustment, if any, could encourage players to consider alternative strategies?

Final Evaluation and Revision

After analyzing the three scenarios, evaluate your redesigned game as a whole.

- Identify **one potential balancing problem** that may remain in your design or emerge because of your changes.
- Propose **one additional adjustment** to address this problem.
- Explain which mechanic you would change, why you would change it, and how you expect the adjustment to affect fairness, challenge, or meaningful choices.

You do not need to rewrite your complete rules or implement this final adjustment.

Remember that your conclusions are based on theoretical analysis. Actual playtesting would be necessary to determine whether your redesigned mechanics function as intended.

Submission

Each student must submit their own copy of the completed activity to Brightspace by 11:59 PM on Thursday, October 1.

Your submission should include:

- **Your name**
- **Your group members' names**
- **Part 1: Initial Analysis**, including three balancing problems
- **Part 2: Investigating Mechanics**, including your calculations, completed table, and written responses
- **Part 3: Redesign**, including your revised action table, balancing explanations, and complete rules
- **Part 4: Triangularity**, including your completed risk-and-reward table and written responses
- **Part 5: Evaluation**, including responses to all three scenarios and one proposed additional adjustment

Important: This is a **group discussion activity**, but submissions are **individual**. You may discuss the game and possible changes with your group, but each student must develop their own analysis and redesign, submit their own copy, and ensure it is complete.

Deadline: Thursday, October 1, by **11:59 PM**.